

ASSIGNMENT – 6

Name: A.V. Prasad

Roll_no:2503A52400

Batch_no:14

Task-1: (Classes-Data Validation)

Prompt:

"Create a Python Student class with a constructor that has three attributes: name, roll _ no, and marks. Add a method called is _pass () that checks whether the student has passed. The student should pass if marks are greater than or equal to 40. Include sample objects and display their details and pass/fail status. Provide clear and simple Python code with an explanation."

Input:

```
class Student:
    def __init__(self, name, roll_no, marks):
        self.name = name
        self.roll_no = roll_no
        self.marks = marks

    def is_pass(self):
        return self.marks >= 40

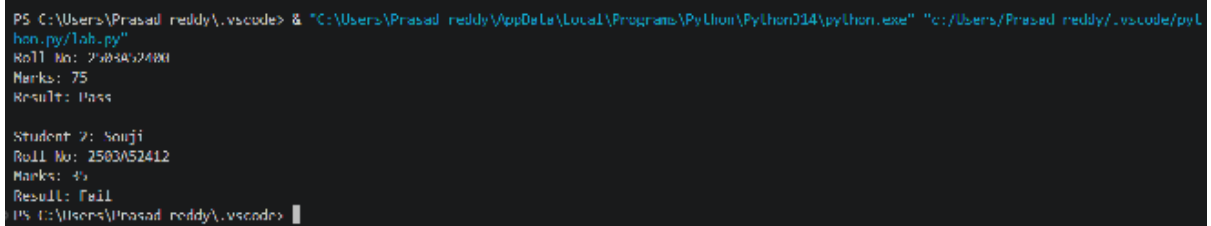
# Creating student objects
student1 = Student("Prasad", "2503A52400", 75)
student2 = Student("Souji", "2503A52412", 35)

print("Student 1:", student1.name)
print("Roll No:", student1.roll_no)
print("Marks:", student1.marks)
print("Result:", "Pass" if student1.is_pass() else "Fail")

print("\nStudent 2:", student2.name)
print("Roll No:", student2.roll_no)
print("Marks:", student2.marks)
```

```
print("Result:", "Pass" if student2.is_pass() else "Fail")
```

Output:



```
PS C:\Users\Prasad_reddy\.vscode> & "C:\Users\Prasad_reddy\AppData\Local\Programs\Python\Python314\python.exe" "c:/Users/Prasad_reddy/.vscode/python.py/lab.py"
Roll No: 2503A52400
Marks: 75
Result: Pass

Student 2: Souji
Roll No: 2503A52412
Marks: 45
Result: Fail
PS C:\Users\Prasad_reddy\.vscode>
```

Task-2:(Loops: Pattern Generation)

Prompt:

"Write a Python function that generates and prints a right-angled triangle star pattern using a for and while loop. The function should accept the number of rows as input. For example, if the number of rows is 5, the output should contain 1 star in the first row, 2 stars in the second row, and continue until 5 stars in the fifth row. Provide the code and explain how the loop works."

Input:

```
def triangle_for(n):
    for i in range(1, n + 1):
        print("*" * i)
```

```
print("using for loop:")
triangle_for(5)
```

```
def triangle_while(n):
    i = 1
```

```
    while i <= n:
        print("*" * i)
        i += 1
```

```
print("using while loop:")
```

```
triangle_while(5)
```

Output:

```
PS C:\Users\Prasad reddy\.vscode> & "C:\Users\Prasad reddy\AppData\Local\Programs\Python\Python314\python.exe" "c:/Users/Prasad reddy/.vscode/python.py/lab.py"
using for loop:
*
**
***
****
*****
using while loop:
*
**
***
****
*****
PS C:\Users\Prasad reddy\.vscode> 
```

Task-3:(Conditional Statements: Number Analysis)

Prompt:

"Write a Python function called check _ number(num) that determines whether a given number is positive, negative, or zero using if-elif -else statements. Test the function with multiple values, including a positive number, a negative number, and zero. Display the results clearly and explain the decision-making logic."

Input:

```
def check_number(num):
```

```
    if num > 0:
```

```
        return "Positive"
```

```
    elif num < 0:
```

```
        return "Negative"
```

```
else:
```

```
    return "Zero"
```

```
# Testing multiple inputs
```

```
numbers = [25, -10, 0, 7, -3]
```

```
for number in numbers:
```

```
    print(number, ":", check_number(number))
```

Output:

```
PS C:\Users\Prasad reddy\.vscode> & "C:\Users\Prasad reddy\AppData\Local\Programs\Python\Python314\python.exe" "c:/Users/Prasad reddy/.vscode/pyt
hon.py/lab.py"
25 : Positive
-10 : Negative
0 : Zero
7 : Positive
-3 : Negative
PS C:\Users\Prasad reddy\.vscode>
```

Task-4:(Nested Conditionals: Discount Eligibility)

Prompt:

"Write a Python function called `check _ discount (age, is_ member)` using nested if statements. If the person's age is greater than or equal to 60, they should be eligible for a senior discount. If the person is also a member, they should receive an additional member discount. If the person is below 60, check separately whether they are a member and display the appropriate discount eligibility. Test the function with multiple combinations of age and membership status. Explain the decision flow."

Input:

```
def check_discount(age, is_member):
    if age >= 60:
        print("Senior discount eligible")

        if is_member:
            print("Additional member discount")
        else:
            print("Not eligible for member discount")

    else:
        print("Not eligible for senior discount")

        if is_member:
            print("Member discount eligible")
        else:
            print("No discount available")
```

Test cases

```
print("Customer 1:")  
check_discount(65, True)
```

```
print("\nCustomer 2:")  
check_discount(65, False)
```

```
print("\nCustomer 3:")  
check_discount(30, True)
```

```
print("\nCustomer 4:")  
check_discount(30, False)
```

Output:

```
PS C:\Users\Prasad_reddy\.vscode> & "C:\Users\Prasad_reddy\AppData\Local\Programs\Python\Python314\python.exe" "C:\Users\Prasad_reddy\.vscode\python.py/lab.py"  
Customer 1:  
Senior discount eligible  
Additional member discount  
  
Customer 2:  
Senior discount eligible  
Not eligible for member discount  
  
Customer 3:  
Not eligible for senior discount  
Member discount eligible  
  
Customer 4:  
Not eligible for senior discount  
No discount available  
PS C:\Users\Prasad_reddy\.vscode>
```

Task-5:(Class: Circle Mathematical Operations)

Prompt:

"Create a Python Circle class with a constructor that accepts radius as an attribute. Add one method called area () to calculate the area of the circle and another method called circumference () to calculate its circumference. Use the formulas $\text{Area} = \pi \times r^2$ and $\text{Circumference} = 2 \times \pi \times r$. Create a Circle object with a sample radius, calculate both values, display the results, and provide a clear explanation of the code."

Input:

```
import math

class Circle:
    def __init__(self, radius):
        self.radius = radius

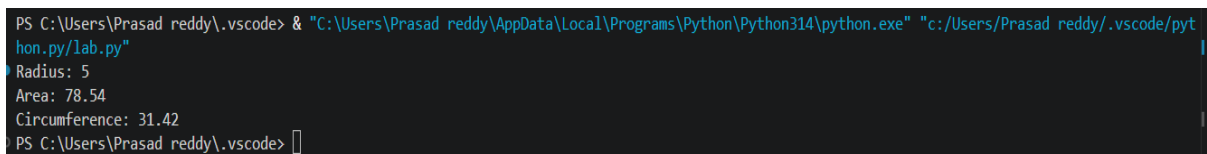
    def area(self):
        return math.pi * self.radius ** 2

    def circumference(self):
        return 2 * math.pi * self.radius

# Creating a Circle object
circle = Circle(5)

print("Radius:", circle.radius)
print("Area:", round(circle.area(), 2))
print("Circumference:", round(circle.circumference(), 2))
```

Output:



```
PS C:\Users\Prasad reddy\.vscode> & "C:\Users\Prasad reddy\AppData\Local\Programs\Python\Python314\python.exe" "c:/Users/Prasad reddy/.vscode/python.py/lab.py"
Radius: 5
Area: 78.54
Circumference: 31.42
PS C:\Users\Prasad reddy\.vscode>
```

